MIL-H-3199E
21 January 1975
SUPERSEDING
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1 October 1969

MILITARY SPECIFICATION

HEATERS, VEHICULAR COMPARTMENT
28 VOLT, DC

This specification is approved for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heaters, of 28-volt DC nominal rating, used for heating crew and engine compartments of military vehicles (see 6.1).

1.2 Classification. Heaters shall be of the following types and capacities, as specified (see 6.2):

Type I - Gasoline Fired:
  Capacity A - 20,000 BTU/HR.
  Capacity B - 30,000 BTU/HR.
  Capacity C - 60,000 BTU/HR.

Type II - Liquid Coolant Operated:
  Capacity A - 20,000 BTU/HR.
  Capacity B - 30,000 BTU/HR.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC-2540

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SPECIFICATIONS

Federal

O-A-548	- Antifreeze, Ethylene Glycol, Inhibited.
O-I-490	- Inhibitor, Corrosion, Liquid Cooling System.
VV-G-76	- Gasoline, Automotive.

Military

MIL-P-514	- Plates; Identification, Instruction and Marking, Blank.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-C-11755	- Compound, Antifreeze, Arctic-Type.
MIL-P-14105	- Paint, Heat-Resisting, Olive Drab (for Steel Surfaces).

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-Identification Marking of U. S. Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirement for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MS-51326	- Heater, Vehicular Compartment, Hot Water, 30,000 BTU/HR.
MS-51327	- Heater, Vehicular Compartment, How Water, 20,000 BTU/HR.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. DEPARTMENT OF COMMERCE
(Publication No. 235 - Weather Bureau Psychrometric Tables.)

U.S. DEPARTMENT OF TRANSPORTATION
(Motor Carrier Safety Regulations (MCSR)

(Application for copies should be addressed to the Superintendent of Documents U. S. Government Printing Office, Washington, D.C., 20402.)

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3. REQUIREMENTS

3.1 First article. A first article of the heaters to be furnished under this specification shall be produced prior to the manufacture of the items in production quantities. Heaters furnished under this specification shall be products which are represented by a first article sample that has been tested and has passed the first article inspection specified herein (see 4.2). Approval of the first article sample by the activities concerned shall not relieve the contractor of his obligation to supply heaters and production that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from the first article sample shall be subject to the approval of the contracting officer.

3.2 Design and construction. Unless otherwise specified, Type I heaters shall fall within the envelope outline of the applicable drawings and shall conform to the detail requirements for mounting dimensions, external electrical connections and fuel inlet. Type II heaters shall be constructed in accordance with MS-51326 or MS-51327 as applicable (see 6.2) Type I and Type II heaters shall meet the applicable requirements of Section 393.77, MCSR in effect on date of manufacture.

3.2.1 Interchangeability. All parts having the same manufacturer's or Army part numbers shall be completely interchangeable with each other with respect to installation and performance.

3.3 Controls. When specified (see 6.2), heaters shall be furnished complete with controls to start, stop, and operate heater.

3.4 Coolants. Type II heaters shall perform as specified herein using a water and antifreeze compound conforming to O-A-548, arctic-type antifreeze compound conforming to MIL-C-11755, or water and corrosion inhibitor compound conforming to O-I-490.

3.5 Fuel. Type I heaters shall perform as specified in 3.6.1, using gasoline conforming to MIL-G-3056 or VV-G-76, as applicable.

3.6 Performance.

3.6.1 Type I heaters.

3.6.1.1 Heat output. When tested as specified in 4.5.2.1, the type I heaters shall have a heat output as specified in table I.

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Table I. Heat output (type I)

Capacity	BTU per hour (Minimum)	
	High setting	Low setting
A	18,000	7,200
B	27,000	16,200
C	54,000	27,000

3.6.1.2 Air discharge rate. During test specified in 4.5.2.1, the heaters when performing at high setting, shall discharge air at not less than the rates specified in table II.

Table II. Air discharge (type I)

Capacity	Cubic feet per minute	
	Unrestricted	Air back pressure
A	80	50 at 1/2 inch H_2O
B	105	85 at 1 inch H_2O
C	200	150 at 1 inch H_2O

3.6.1.3 Fuel flow. When tested as specified in 4.5.2.2, using fuel conforming to VV-G-76 or MIL-G-3056 as applicable, the flow rate shall be not more than the values specified in table III.

Table III. Fuel flow (type I)

Capacity	Gallons per hour	
	High heat	Low heat
A	.29	.13
B	.48	.30
C	.93	.44

3.6.1.4 Current. When tested as specified in 4.5.2.3, the current shall be not more than the values specified in table IV.

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Table IV. Current

Capacity	Amperes	
	Start	Run
A	12	4
B	18	7
C	21	11

3.6.1.5 Overheat switch. When tested as specified in 4.5.2.4, the overheat switch shall break the circuit and cut off the flow of fuel when the ventilating air mixed with ambient air is heated to a temperature of 375° plus or minus 25°F

3.6.1.6 Cycling. When tested as specified in 4.5.2.5, the heaters shall start in less than four minutes and purge in less than three minutes.

3.6.1.7 Operating position. When tested as specified in 4.5.2.6, heaters shall start and operate without difficulty.

3.6.1.8 Endurance. When tested as specified in 4.5.2.7, the heater shall operate without a major failure and conform with 3.6.1.1, 3.6.1.2, 3.6.1.3, and 3.6.1.4. A major failure is defined as a breakdown of the heat exchanger, electrical motor, combustion air blower wheel, ventilating air blower wheel, or fuel control value.

3.6.2 Type II heaters.

3.6.2.1 Heat output and air discharge. When tested as specified in 4.5.3.1, while operating in any plane, heat output and air discharge rate shall not be less than amounts shown in table V.

Table V. Heat output and air discharge (type II)

Capacity	Inches H_2O back pressure at air outlet	Heat output BTU/HR*	Air discharge CFM standard conditions
A	0	20,000	150
	1.6	12,000	75
B	0	28,000	150
	1.6	15,000	75

3.6.2.2 Cycling. When tested as specified in 4.5.3.2, heater shall have no coolant leakage from the coolant circulating system.

3.6.2.3 Endurance. When tested as specified in 4.5.3.3, the heater shall operate without a breakdown of the heat exchanger, electric motor, or fan, and shall conform to 3.6.2.1. Starting time to full load speed shall be 10 seconds maximum.

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3.6.3 Type I and II heaters.

3.6.3.1 Voltage. The heater shall start and operate satisfactorily at an applied voltage of 20 to 28.5 volts DC.

3.7 Environmental.

3.7.1 Type I heaters.

3.7.1.1 Low temperature operation. When tested as specified in 4.5.4.1.1, the heaters shall have operational stability while conforming to 3.6.1.1.

3.7.1.2 Vibration. When tested as specified 4.5.1.2, the heaters shall start and operate while vibrating. After testing, heater shall have no evidence of fuel leakage and shall meet the requirements of 3.6.1.1.

3.7.2 Type II heaters.

3.7.2.1 Vibration. When tested as specified in 4.5.4.2.1, the heaters shall start and operate while vibrating. After test, heaters shall have no evidence of coolant leakage and shall meet the requirements of 3.6.2.1.

3.7.3 Type I and II heaters.

3.7.3.1 Corrosion resistance. When tested as specified in 4.5.4.3.1, the heaters shall show no evidence of corrosion that would affect their functional operation after being subjected to a 96-hour salt solution spray.

3.7.3.2 Electromagnetic interference. When tested as specified in 4.5.4.3.2, the heater shall meet the electromagnetic interference requirements of MIL-STD-461 for class IIC equipment.

3.8 Painting.

3.8.1 Type I heaters. The exterior surfaces of the heater case and outlet adapter shall be cleaned, degreased, and then painted with paint conforming to MIL-P-14105. Other surfaces of the heater shall be finished in accordance with MIL-STD-193.

3.8.2 Type II heaters. The exterior surfaces of the heater shall be finished in accordance with MIL-STD-193.

3.9 Indentification of product.

3.9.1 Marking. The heater shall be marked in accordance with MIL-STD-130.

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3.9.2 Nameplate. A nameplate shall be attached to the heater in a location, so that the plate may be easily read when the heater is installed in the vehicle. The nameplate shall conform to MIL-P-514 and shall contain the following minimum information:

Heater 28 volts DC
Military part number
Federal stock number
Manufacturer's identification
Manufacturer's serial number
US

3.10 Workmanship. Workmanship shall be such as to assure a product free of rust, burrs, and scratches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. The first article sample (see 3.1) shall consist of two heaters of each type and capacity. The specimens, properly marked with identifying information, shall be representative of the units proposed to be furnished to the Government. Unless otherwise specified, first article inspection shall be done by the manufacturer under Government surveilance and shall consist of examination for the defects specified in 4.3.2.2 and testing as specified in table VI.

Table VI. Order of first article testing

Specimen	Type I Par No.	Heater Test	Type II Par No.	Heater Test
1	4.5.2.1	Heat output and air discharge	4.5.3.1	Heat output and air discharge
	4.5.2.2	Fuel flow	4.5.3.2	Cycling
	4.5.2.3	Current		
	4.5.2.5	Cycling	4.5.4.3.1	Corrosion
	4.5.2.6	Operating position	4.5.4.3.2	Electromagnetic Compatibility
	4.5.4.3.2	Electromagnetic compatibility		

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Table VI. Order of first article testing (Cont'd)

Specimen	Type I Par No.	Heater Test	Type II Par No.	Heater Test
2	4.5.4.1.2	Vibration	4.5.4.2.1	Vibration
	4.5.2.4	Overheat	4.5.3.3	Endurance
	4.5.2.7	Endurance		
	4.5.4.3.1	Corrosion		
	4.5.2.1	Heat output and air discharge		

4.2.1 Failure. Failure of the heater to pass any of the examinations or tests specified herein may be cause for the Government to refuse to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling for heater assemblies.

4.3.1.1 Lot formation. A lot shall consist of all heaters from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for testing. Samples for testing shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each heater selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as shown in table VII.

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Table VII. Classification of defects

Categories	Defects
Critical	None
Major:	
101	Dimensions affecting interchangeability not within tolerance (see 3.2.1).
102	Controls, missing (see 3.3).
Minor:	
201	Dimensions not affecting interchangeability not within tolerance (see 3.2.1).
202	Improper finish (see 3.8).
203	Improper marking (see 3.9).
204	Poor workmanship (see 3.10).

4.3.3 Testing.

4.3.3.1 Testing. Heaters selected in accordance with 4.3.1.3, shall be subjected to the tests specified in table VIII using an AQL of 6.5 on the basis of percent defective.

Table VIII. Order of testing

Test	Type I Par. No.	Type II Par. No.
Heat output and air discharge	4.5.2.1	4.5.3.1

4.4 Control tests.

4.4.1 Sampling. Control test samples shall be selected at the rate of one heater for every 100 produced (or portion thereof), except that not more than three heaters shall be selected in any 30 day period. Samples shall be from a production lot which has passed the examination specified in 4.3.2 and the testing specified in table VIII.

4.4.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.5 Test procedures.

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4.5.1 General.

4.5.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. The ambient room temperature of the laboratory shall be 77° ± 15°F. Unless otherwise specified herein or in applicable specifications, specimens shall be tested at the ambient room temperature of the laboratory. Heaters shall be heat stabilized for a period of one hour prior to being subjected to any test excepting that the one hour stabilization period shall be omitted for consecutive tests at the same ambient temperature. Tests results shall be corrected to standard sea level conditions. Unless otherwise specified herein, the test voltage shall be 28 ± .5 volts.

4.5.1.2 Calibration of test equipment. Unless otherwise specified herein, test equipment accuracy shall be as to permit measurement of 10 percent of product on test specification tolerance. Calibration of test equipment shall be conducted at intervals to establish required accuracy. Records of calibration shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government. Required measuring instrument tolerances are as listed in table IX.

Table IX. Measuring instrument tolerances

Instrument	Accuracy of tolerance
Ammeter DC	.50 percent
Voltmeter DC	.50 percent
Temperature gage	1.00 percent

4.5.2 Test procedures, type I heaters.

4.5.2.1 Heat output and air discharge.

4.5.2.1.1 Heat output equipment. The test equipment shall consist essentially of a plenum chamber, outlet duct, thermocouples, and nozzle; set-up to measure the air temperature rise and air flow through the heater.

4.5.2.1.2 Temperature. Thermocouples shall be installed at the heater air inlet, and the inlet air temperature shall be minus 65° plus or minus 5°F. The heater air outlet temperature shall be determined by a series to thermocouples located in the plenum chamber and at the nozzle. The air temperature rise through the heater shall be the temperature difference of the entering and leaving air.

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4.5.2.1.3 Heat output. The output of the heater shall be determined by the following formula:

$$H = Vd\,C_p\,(T_1 - T_2).$$

H = heat output in BTU per hr.
V = air flow in cu ft per hr.
d = density of air.
C_p = specific heat of air, .24 BTU per lb. per ^{o}F.
T_1 = inlet air temperature in ^{o}F.
T_2 = outlet air temperature in ^{o}F.

Record measurements necessary to calculate heat output to determine conformance to 3.6.1.1.

4.5.2.1.4. Air discharge. The heater outlet air flow shall be measured utilizing air flow measuring equipment and methodology traceable to the National Bureau of Standards. If a pitot tube is utilized, calculate the air discharge using the following formula to determine conformance to 3.6.1.2:

$$Q = 1097\ A\sqrt{\frac{h_v}{d}}$$

where Q = air discharge in cu. ft. per minute
A = cross-sectional area in sq. ft.
h_v = velocity pressure in inches of water
d = density of air in lbs. per cu. ft.

4.5.2.2 Fuel flow. A solenoid controlled three-way valve shall be connected before the fuel pump with one valve inlet connected to a burette and the other inlet to the gasoline supply. The solenoid valves shall be connected electrically to a timing device which will open the burette side of the valve for two minutes. Alternate systems may be used for this test, but must be approved by the responsible activity. The fuel flow shall be checked to determine conformance to 3.6.1.3.

4.5.2.3 Current. The heaters shall be tested utilizing a starting voltage of 24.0 ± 0.5 volts DC and a test ambient temperature of -65° F. Using the applicable air back pressure restriction on the fresh air outlet, as specified in Table II, the heater shall be started at high heat setting with the heater in the horizontal operating position, operated for 15 minutes and purged. After cooling for 35 minutes using an auxiliary blower, the above procedure shall be repeated on low heat. A total of six (6) runs shall be made, three at high heat and three at low heat, to obtain current values and to determine conformance to 3.6.1.4.

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4.5.2.4 Overheat switch. To determine conformance to 3.6.1.5, the test shall employ a venturi, thermocouple, and ducts. The heater shall be mounted to exhaust ventilating air through a duct of approximately five feet in length. A venturi and thermocouples shall be placed at the duct. The heater shall be run on high heat and ventilating air flow restricted by holding a piece of cardboard or sheet metal over the duct or venturi outlet. When shutting off ventilating air supply, care must be taken to insure exhaust air supply.

4.5.2.5 Cycling. To determine conformance to 3.6.1.6, the heater in normal operating position, shall be cycled at high setting for 50 cycles at an ambient temperature of minus 65° plus or minus 5°F. Each cycle shall consist of starting at high heat, running for 15 minutes, purging, and cooling the heater for 35 minutes with auxiliary blower. The start time shall be measured from the time the heater is turned on until the igniter circuit is de-energized. The purge time shall be measured from the time the heater switch is turned off until the time blowers shut off. The average starting and purge time shall be recorded and shall not exceed 4.0 and 3.5 minutes respectively.

4.5.2.6 Operating position. To determine conformance with 3.6.1.7, the heater shall be started and run for 45 minutes in each of the tilted and rotated positions specified herein. The blower intake end shall be on or above the horizontal plane. The test shall be conducted in the following positions:

Longitudinal axis:	Rotated position:
(a) Horizontal	Normal operating
(b) Horizontal	90° turn
(c) Horizontal	180° turn
(d) 45°	Base downward
(e) 45°	90° turn
(f) 45°	180° turn
(g) Vertical	

4.5.2.7 Endurance. To determine conformance to 3.6.1.8, the heater shall be mounted in the normal horizontal position. The heater shall be started and run for 125 hours at each ambient temperature of 0°F, -25°F, -45°F, and -65°F, for a total of 500 hours, operating alternately for 24 hours at high and low heat settings while cycling 20 minutes "ON" and 10 minutes "OFF". "OFF" time shall include purge time. The heater shall be operated 50 percent of the time at 20 ± 0.5 volts, 25 percent at 24 ± 0.5 volts, and 25 percent at 28 ± 0.5 volts. Use "ON" time only for totalling the 500 hours. The performance of the heater shall be checked at the beginning, midway and at the end of the test for conformance to 3.6.1.8.

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4.5.3 Test procedures, type II heaters.

4.5.3.1 Heat output and air discharge. The heater shall be tested in an ambient temperature of minus 25° plus or minus 5°F and the coolant shall circulate through the core at a rate of 30 pounds plus or minus two pounds per minute. Temperature of the coolant entering the heater inlet shall be maintained at plus 180° plus or minus 5°F. Heat output and air discharge of the heater shall be measured by a venturi equipped with thermocouples to determine conformance to 3.6.2.1.

4.5.3.2 Cycling. The coolant, circulating through the system, shall be subjected to a pressure rise and fall from 0 to to 32 psi at the rate of 20 cycles per minute in an ambient temperature of minus 25° plus or minus 5°F. After 2,500 cycles, the heater shall be checked for coolant leakage to determine conformance with 3.6.2.1.

4.5.3.3 Endurance. The heater shall be set up as in 4.5.2.1. The temperature of the coolant to the inlet shall be maintained at plus 180° plus or minus 5°F. The heater shall be operated for 250 hours at 0 back pressure and 250 hours at 1.6 inches of water back pressure to determine conformance to 3.6.2.3. At the beginning, midway, and at end of test, the heater shall be checked to determine conformance to 3.6.2.1.

4.5.4 Environmental.

4.5.4.1 Type I heaters.

4.5.4.1.1 Low temperature. To determine conformance to 3.7.1.1, the heater shall be run for 100 hours at minus 65° plussor minus 5°F on alternate high and low heat settings with total of 70 hours on high heat.

4.5.4.1.2 Vibration. The heater shall be vibrated at an amplitude of 0.025 inch for one hour in the direction of each of the three major axis of the heater. While accelerating and decelerating at a uniform rate for an interval of one minute, the frequency shall vary through a range of 10 to 55 cycles per second. After 10 minutes of vibrating, the heater shall be started, run for 20 minutes, and then shut off (but vibrating continued); this procedure shall be followed for each axis. After vibrating on each respective axis, the heater shall be checked for fuel leaks, to determine conformance with 3.7.1.2.

4.5.4.2 Type II heaters.

4.5.4.2.1 Vibration. The heater at a test temperature of minus 25°F shall be vibrated at an amplitude of 0.025 inch for two hours in the direction of each of the three major axis of the heater. While accelerating and decelerating at a uniform rate for an interval of one minute, the frequency shall vary through a range of 10 to 55 cycles per second. After 40 minutes of

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vibrating, the heater blower shall be started, run for one hour and then shut off (but vibrating continued); this procedure shall be followed for each axis. After vibrating, the heater shall be checked for conformance to 3.7.2.1.

4.5.4.3 Type I and II heaters.

4.5.4.3.1 Corrosion. Heaters, with all duct openings sealed, shall be subjected to a salt spray corrosion test method 101 of MIL-STD-202, to determine conformance to 3.7.3.1.

4.5.4.3.2 Electromagnetic interference. To determine conformance to 3.7.3.2, the heater shall be tested in accordance with the applicable provisions of MIL-STD-462.

4.6 Inspection of preparation for delivery.

4.6.1 Materials and processes. All materials and processes involved in the preparation for delivery shall be inspected to determine conformance to requirements of Section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use.

6.1.1 Type I heaters. Type I heaters are intended for use on military vehicles or equipment which may be required to operate in an ambient temperature to minus 65°F; from which a supply of gasoline is available; and when heated air is required without the vehicle engine being operated.

6.1.2 Type II heaters. Type II heaters are intended for use on military vehicles or equipment from which a supply of heated circulating liquid (engine coolant) is available, and vehicles which may be required to operate in any ambient temperature to minus 25°F.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and capacity (see 1.2).
(c) Applicable Military Standard or drawing (see 3.2)

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(d) If controls are to be furnished (see 3.3).
(e) Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Derivation of formula. The formula noted in 4.5.2.1.4 was derived as follows:

6.3.1 Head. The head in feet of air, is obtained as follows:

$$h = \frac{62.31\ p}{12\ d} = \frac{5.19p}{d}$$

h = head in feet of air.
d = density of air in lbs per cu ft.
p = pressure or head in inches of water.
62.31 = density of water in lbs per cu ft at 70°F.

6.3.2 Velocity. Velocity of air flow in ft per min. is obtained as follows:

$$V = 60\sqrt{2gh}$$
$$= 60\sqrt{\frac{2 \times 32.2 \times 5.19\ p}{d}}$$
$$= 1096.9\sqrt{\frac{p}{d}}$$

Custodians:
Army - AT
Air Force - 85

Review Activity:
DSA - CS

User activity:
Navy - MC
Navy - YD

Preparing activity:
Army - AT

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

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